



Machine Problem No. 1			
Topic:	Fundamentals of Machine Learning	Week No.	2
Course Code:	CSST102	Term:	1 st Semester
Course Title:	Basic Machine Learning	Academic Year:	2025-2026
Student Name		Section	
Due date		Points	

Fundamentals of Machine Learning

Topic: What is ML? Types of ML and Core Challenges

Objectives

By the end of the lab, students should be able to:

1. Load and explore a dataset using Scikit-Learn.
2. Perform a train-test split to prepare data.
3. Train a simple baseline ML model.
4. Evaluate model performance using metrics.
5. Relate the task to supervised vs. unsupervised learning.

Lab Outline (3 hours)

Hour 1 – Setup & Dataset Exploration

- Install/verify Python, Jupyter/Colab, and Scikit-Learn.
- Load the **Iris dataset** (classification) or **California Housing dataset** (regression).
- Explore dataset (features, targets, summary statistics).



Code Snippet:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df=pd.read_csv("iris_dataset.csv")
print(df.head(5))

print(df.info)
print(df.describe())
print(df['species'].value_counts())

df['sepal length (cm)'].hist()
plt.title("Sepal Length Distribution")
plt.show()

sns.scatterplot(data=df,x="petal length (cm)",y='petal width (cm)',hue="species")
plt.title("Petal Length vs Width by Species")
plt.show()
```

Mini-task: Students answer:

- What is the **input (features)**?
 - What is the **output (label)**?
 - Is this **supervised or unsupervised learning**?
-

Hour 2 – Train-Test Split & Baseline Model

- Perform train-test split (80% train, 20% test).
- Train a simple baseline model:
 - Logistic Regression (for Iris)
 - Linear Regression (for Housing)
- Make predictions.



Code Snippet:

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Features and target
X = df.drop("species", axis=1)
y = df["species"]

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train Decision Tree
clf = DecisionTreeClassifier(random_state=42)
clf.fit(X_train, y_train)

# Predictions
y_pred = clf.predict(X_test)
```

Mini-task: Students compute model accuracy.

Hour 3 – Evaluation & Reflection

- Evaluate model with different metrics:
 - Classification: Confusion matrix, precision, recall.
 - Regression: RMSE (Root Mean Squared Error).
- Discuss ML challenges: overfitting, underfitting, and bad data.
- Students reflect:
 - “What would happen if the dataset had missing or wrong values?”
 - “How does this relate to real-world ML applications?”

Code Snippet (Confusion Matrix):

```
# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion matrix
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```



Deliverables (Lab Submission)

1. Python notebook (Jupyter/Colab) with:
 - Dataset loading & exploration
 - Train-test split
 - Model training & evaluation
2. Short reflection (3–5 sentences):
 - What ML type did you use?
 - What challenge might affect the model?

Assessment (30 points)

- Dataset Exploration (5 pts)
- Train-Test Split (5 pts)
- Baseline Model Training (10 pts)
- Evaluation Metrics (5 pts)
- Reflection/Discussion (5 pts)

 **Total: 30 points**